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Question Paper Code : 20645

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2023

Third Semester

Computer Science and Engineering

19CS391 – PROGRAMMING WITH JAVA

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

U, CO1 1. Identify modifiers used for methods in Java.

A, CO1 2. Predict the output of given Java program: Justify your answer.

```
class Test {
public static void swap(Integer i, Integer j) {
    Integer temp = new Integer(i);
    i = j;
    j = temp;    }
  public static void main(String[] args) {
    Integer i = new Integer(10);
    Integer j = new Integer(20);
    swap(i, j);
    System.out.println("i = " + i + ", j = " + j);    } }
```

A) i = 10, j = 20 B) i = 20, j = 10 C) i = 10, j = 10 D) i = 20, j = 20

U, CO2 3. What is the purpose of abstract classes and methods in Java?

A, CO2 4. What will be the output of the following Java program? Justify your answer.

```
class exception_handling{
        public static void main(String args[]){
                try{
                        System.out.print("Hello" + " " + 1 / 0); }
                catch(ArithmeticException e){
                        System.out.print("World"); }          }       }
```

A) Hello B) World C) HelloWorld D) Hello World

U, CO4 5. What are the main objectives of the native keyword in Java?

A, CO3 6. What will be the output of the following Java code snippet? Justify your answer.

```
public class Demo{
        public static void main(String[] args){
                Map<Integer,Object> sampleMap=new TreeMap<Integer, Object>();
                sampleMap.put(1, null);
                sampleMap.put(5, null);
                sampleMap.put(3, null);
                sampleMap.put(2, null);
                sampleMap.put(4, null);
                System.out.println(sampleMap);      }    }
```

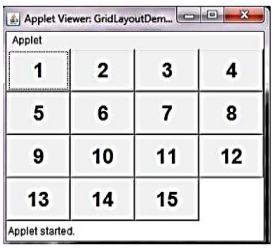
A) {1=null, 2=null, 3=null, 4=null, 5=null} B) {5=null}

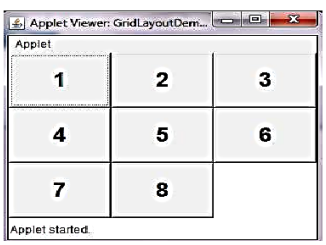
C) {1=null, 5=null, 3=null, 2=null, 4=null} D) Exception is thrown

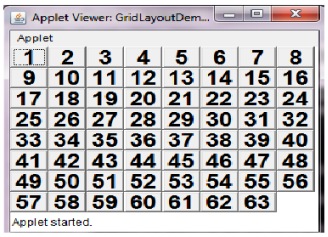
R, CO5 7. List any four AWT event Adapter classes in Java along with its interfaces.

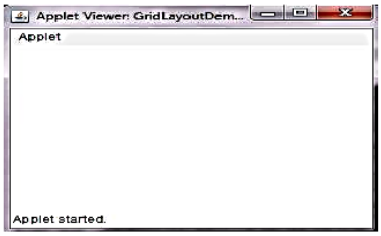
A, CO5 8. What will be the output for following code? Justify your answer.

```
import java.awt.*;
import java.applet.*;
/*<applet code="GridLayoutDemo11" width=300 height=200> </applet> */
public class GridLayoutDemo11 extends Applet {
        static final int n = 4;
        public void init() {
                setLayout(new GridLayout(n, n));
                setFont(new Font("Sansserif", Font.BOLD, 24));
                for(int i = 0; i < n; i++) {
                        for(int j = 0; j < n; j++) {
                                int k = i * n + j;
                                if(k > 0)
                                        add(new Button("" + k)); } } } }
```

A) 

B) 

C) 

D) 

U, CO6 9. Why do we need thread synchronization?

U, CO6 10. Number of threads in below Java program is _____. Justify your answer.

```
public class ThreadExtended extends Thread {
        public void run() {
                System.out.println("Thread is running no");  }
        public static void main(String[] args){
                ThreadExtended threadE = new ThreadExtended();
                threadE.start();        }       }
```

A) 0 B) 1 C) 2 D) 3

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. Explain the purpose of static keyword in various aspects of Java with an example. (8 Marks)

A, CO1 11. a) ii. Implement a Java program to define a class called 'Mobilephone' with Model Number, Manufacture Name, Cost, Size, Color, MemoryCapacity, CameraPixelSize, etc.,. Include various types of constructors such as Default constructor, Parameterized constructor with two arguments and three arguments. Create another class to write main for Mobilephone class that initiate array of objects to invoke various types of constructors. (8 Marks)

OR

U, CO1 11. b) i. Explain string Tokennizer class statements in Java with an example. (8 Marks)

A, CO1 11. b) ii. Write a Java program to perform binary search on key = 11 on the following list = {2, 4, 7, 10, 11, 45, 50, 59, 60, 66, 69, 70, 79} (8 Marks)

A, CO2 12. a) Create a package called shape2d. In that package declare an interface called TWO_D and three classes Square, Rectangle and Circle. In all three classes implement the twod interface. In

interface twod declare two methods area() and perimeter(). Both returns double value. Square class contains one instance variable called side of double type, a parameterized constructor for initializing that member, area() and perimeter() methods for computing and returning the area and perimeter of the square using appropriate formulae. Similarly Rectangle class contains two instance variables called length and breadth of double type, a parameterized constructor with two parameters for initializing those members, area() and perimeter() methods for computing and returning the area and perimeter of the rectangle using appropriate formulae. Circle class contains one instance variable called radius of double type, a parameterized constructor for initializing that member, area() and perimeter() methods for computing and returning the area and perimeter of the circle using appropriate formulae. Test the program polymorphically. (16 Marks)

OR

A, CO2 12. b) Create a Point class that has two data members x and y of double type. Write a default constructor which initializes its data members to zero. Write a parameterized constructor which takes two parameters of double type and assigns them to its data members. Write another parameterized constructor that takes one Point object as parameter and copies the values of the passed object's data member to the calling object's data members. Write a find_distance method that takes two double parameters representing the x, y values of a point and finds the distance between the calling object and the passed parameters and returns the distance as double value. Overload the find_distance method, that takes single Point object parameter and computes the distance between both the points and returns the distance as a double value. Write a user defined Java exception to display "GIVE VALID POINTS" if point values are zero. Write a display method to print the point in the format of "(x, y)". In main method create three point objects p1, p2, p3 and p4. Initialize p1 with (3.25, 7.89), p2 with (5.37, 18.12) and p3 with p2. Find distance between p1 and (7.9, 16.25) using first

find_distance method and between p1 and p3 using second find_distance method. Finally initialize p4 with (0.0, 0.0) and print the results. (16 Marks)

U, CO3 13. a) i. Why do we go for generic programming? Justify your answer with example generic classes and methods in Java. (8 Marks)

A, CO3 13. a) ii. Write a Java program to create an array list, add some colors (strings) and do the following:

A) Iterate through all elements in an array list and print the collection
B) Insert a new color into the array list at the first position
C) Sort in ascending order and print the array list
D) Reverse elements in an array list (8 Marks)

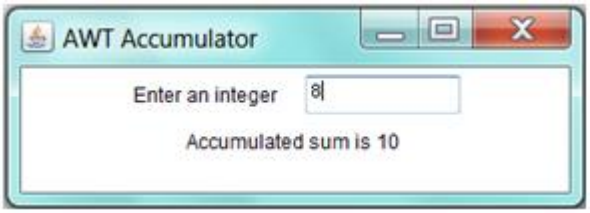
OR

U, CO3 13. b) i. Explain the need for object serialization in Java with a sample program. (8 Marks)

A, CO4 13. b) ii. Write a Java program to associate the (key, value) pair that represents employee (id, name) in a HashMap and perform the following:

A) Count the number of key-value (size) mappings in a map
B) Copy all mappings from the specified map to another map
C) Test if a map contains a mapping for the specified key or specified value
D) Remove all mappings from the map and display (8 Marks)

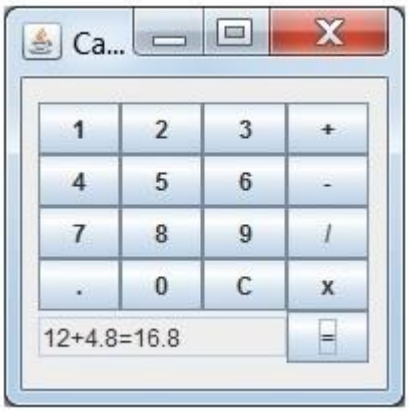
A, CO5 14. a) Write an AWT GUI application called AWTAccumulator, that could calculate accumulated sum of all positive integers entered in a text box similar to diagram given below. When entered integer is zero or negative give an alert message.



(16 Marks)

OR

A, CO5 14. b) Design a simple calculator using Java swings similar to the one given below:



(16 Marks)

U, CO6 15. a) i. Discuss thread states and lifecycle with a neat diagram. (8 Marks)

A, CO6 15. a) ii. Write a Java program that creates two threads to find and print even and odd numbers from 1 to 20. (8 Marks)

OR

U, CO6 15. b) i. Discuss inter thread communication in Java with an example program. (8 Marks)

A, CO6 15. b) ii. Write a Java program that creates a bank account with concurrent deposits and withdrawals using threads. (8 Marks)